MILITARY SPECIFICATION

# MICROCIRCUITS, DIGITAL, ADVANCED SCHOTTKY TTL, SHIFT REGISTERS, MONOLITHIC SILICON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, advanced Schottky TTL, shift register microcircuits. Two product assurance classes and a choice of case outlines and lead finishes are provided and are reflected in the part number.

1.2 Part number. The part number shall be in accordance with MIL-M-38510, and as specified herein.

1.2.1 Device type. The device type shall be as follows:

| Device type | Circuit |
|---|---|
| 01 | 4-bit bidirectional universal shift register |

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outline. The case outline shall be designated as follows:

| Letter | Case outline (see MIL-M-38510, appendix C) |
|---|---|
| E | D-2 (16-lead, 1/4" x 7/8") dual-in-line package |
| F | F-5 (16-lead, 1/4" x 3/8") flat package |
| X | C-2 (20-terminal, .350" x .350") square chip carrier package |

1.3 Absolute maximum ratings.

| | |
|---|---|
| Supply voltage range | -0.5 V to +7.0 V |
| Input voltage range | -1.2 V at -18 mA to +7.0 V |
| Storage temperature range | -65°C to +150°C |
| Maximum power dissipation per device ($P_D$) 1/ | 253 mW |
| Lead temperature (soldering, 10 seconds) | 300°C |
| Thermal resistance, junction to case ($\theta_{JC}$): | |
| Case E | 50°C/W |
| Cases F, and X | 70°C/W |
| Junction temperature ($T_J$) | 175°C |

1/ Must withstand the added $P_D$ due to short-circuit test (e.g., $I_{OS}$).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center (RBE-2), Griffiss AFB, NY 13441, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.



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1.4 Recommended operating conditions.

| | |
|---|---|
| Supply voltage | 4.5 V minimum to 5.5 V maximum |
| Minimum high level input voltage ($V_{IH}$) | 2.0 V |
| Maximum low level input voltage ($V_{IL}$) | 0.8 V |
| Width of clock pulse high $t_{PW(CPH)}$ | 6.0 ns minimum |
| Width of master reset pulse low $t_{PW(MRL)}$ | 5.0 ns minimum |
| Setup time data high to clock pulse $t_{SD(H)}$ | 5.0 ns minimum |
| Setup time data low to clock pulse $t_{SD(L)}$ | 4.0 ns minimum |
| Hold time data high to clock pulse $t_{HD(H)}$ | 2.0 ns minimum |
| Hold time data low to clock pulse $t_{HD(L)}$ | 1.0 ns minimum |
| Setup time select high to clock pulse $t_{SS(H)}$ | 10.0 ns minimum |
| Setup time select low to clock pulse $t_{SS(L)}$ | 8.0 ns minimum |
| Hold time select high to clock pulse $t_{HS(H)}$ | 0.0 ns minimum |
| Hold time select low to clock pulse $t_{HS(L)}$ | 0.0 ns minimum |
| Recovery time master reset to clock pulse $t_{rec(\overline{MR}/CP)}$ | 9.0 ns minimum |
| Input clock frequency $f_{(clock)}$ | 0 - 80 MHz |
| Case operating temperature range ($T_C$) | -55°C to +125°C |

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)



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TABLE I. Electrical performance characteristics.

| Parameter | Symbol | Conditions<br>$-55°C \leq T_C \leq +125°C$ | Device type | Limits | | Unit |
|---|---|---|---|---|---|---|
| | | | | Min | Max | |
| High level output voltage | $V_{OH}$ | $V_{CC}$ = 4.5 V; $I_{OH}$ = -1 mA;<br>$V_{IL}$ = 0.8 V; $V_{IH}$ = 2.0 V | All | 2.5 | | V |
| Low level output voltage | $V_{OL}$ | $V_{CC}$ = 4.5 V; $I_{OL}$ = 20 mA;<br>$V_{IL}$ = 0.8 V; $V_{IH}$ = 2.0 V | All | | 0.5 | V |
| Input clamp voltage | $V_{IC}$ | $V_{CC}$ = 4.5 V; $I_{IN}$ = -18 mA;<br>$T_C$ = +25°C | All | | -1.2 | V |
| High level input current | $I_{IH1}$ | $V_{CC}$ = 5.5 V; $V_{IH}$ = 2.7 V | All | | 20 | µA |
| | $I_{IH2}$ | $V_{CC}$ = 5.5 V; $V_{IH}$ = 7.0 V | All | | 100 | µA |
| Low level input current | $I_{IL1}$ | $V_{CC}$ = 5.5 V; $V_{IL}$ = 0.5 V | All | -.03 | -.60 | mA |
| Short circuit output current 1/ | $I_{OS}$ | $V_{CC}$ = 5.5 V; $V_{OS}$ = 0 V | All | -60 | -150 | mA |
| Output drive test | $I_{OD}$ | $V_{CC}$ = 4.5 V; $V_{IN}$ = 2.5 V | All | 60 | | mA |
| Supply current | $I_{CC}$ | $V_{CC}$ = 5.5 V | All | | 46 | mA |
| Propagation delay time low to high level, CP to Q | $t_{PLH1}$ | $V_{CC}$ = 5.0 V (see figure 5)<br>$C_L$ = 50 pF ±10% | All | 3.0 | 8.5 | ns |
| Propagation delay time high to low level, CP to Q | $t_{PHL1}$ | | All | 3.0 | 8.5 | ns |
| Propagation delay time high to low level, $\overline{MR}$ to Q | $t_{PHL2}$ | | All | 4.5 | 14.5 | ns |

1/ The output conditions have been chosen to produce a current that closely approximates one half of the true short-circuit output current, $I_{OS}$.



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2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510 and herein.

3.2.1 Terminal connections and logic diagram. The terminal connections and logic diagram shall be as specified on figures 1 and 2, respectively.

3.2.2 Truth table. The truth table shall be as specified on figure 3.

3.2.3 Schematic circuits. Schematic circuits shall be submitted to the preparing activity prior to inclusion of a manufacturer's device in this specification and shall be submitted to the qualifying activity as a prerequisite for qualification. All manufacturers' schematics shall be maintained and available upon request.

3.2.4 Case outlines. The case outlines shall be as specified in 1.2.3.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 (see 6.5).

3.4 Electrical performance characteristics. The electrical performance characteristics are as specified in table I, and apply over the full recommended case operating temperature range, unless otherwise specified.

3.5 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

3.6 Marking. Marking shall be in accordance with MIL-M-38510. At the option of the manufacturer, marking of the country of origin may be omitted from the body of the microcircuit but shall be retained on the initial container.

3.7 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 12 (see MIL-M-38510, appendix E).

TABLE II. Electrical test requirements.

| MIL-STD-883 test requirements | Subgroups (see table III) | |
|---|---|---|
| | Class S devices | Class B devices |
| Interim electrical parameters (pre burn-in) (method 5004) | 1 | 1 |
| Final electrical test parameters (method 5004) | 1*,2,3,7,9,10,11 | 1*,2,3,7,9 |
| Group A test requirements (method 5005) | 1,2,3,7,8,9,10,11 | 1,2,3,7,9,10,11 |
| Group B test requirements (method 5005) subgroup 5 | 1,2,3,7,9,10,11 | N/A |
| Group C end-point electrical parameters (method 5005) | N/A | 1,2,3 |
| Additional electrical subgroups for group C periodic inspection | N/A | None |
| Group D end-point electrical parameters (method 5005) | 1,2,3 | 1,2,3 |

* PDA applies to subgroup 1 (see 4.2c).



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4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007, as applicable, of MIL-STD-883, except as modified herein.

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. Burn-in test (method 1015 of MIL-STD-883).

(1) Test condition D or E, using the circuit shown on figure 4 or equivalent.

(2) $T_A$ =+125°C minimum.

b. Interim and final electrical test parameters shall be as specified in table II, except interim electrical parameters test prior to burn-in is optional at the discretion of the manufacturer.

c. The percent defective allowable (PDA) for class S devices shall be as specified in MIL-M-38510. The PDA for class B devices shall be 10 percent based on failures from group A, subgroup 1 test after cooldown as the final electrical test in accordance with method 5004 of MIL-STD-883, and with no intervening electrical measurements. If interim electrical parameter tests are performed prior to burn-in, failures resulting from pre burn-in screening may be excluded from the PDA. If interim electrical parameter tests prior to burn-in are omitted, then all screening failures shall be included in the PDA. The verified failures of group A, subgroup 1, after burn-in divided by the total number of devices submitted for burn-in in that lot shall be used to determine the percent defective for that lot, and the lot shall be accepted or rejected based on the PDA for the applicable device class.

4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Tests shall be as specified in table II herein.

b. Subgroups 4, 5, and 6 of table I of method 5005 of MIL-STD-883 shall be omitted.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883. Electrical parameters shall be as specified in table II herein.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein.

b. Subgroups 3 and 4 shall be added to group C inspection for class B devices, and shall consist of the tests, conditions, and limits specified for subgroups 10 and 11 of group A inspection.



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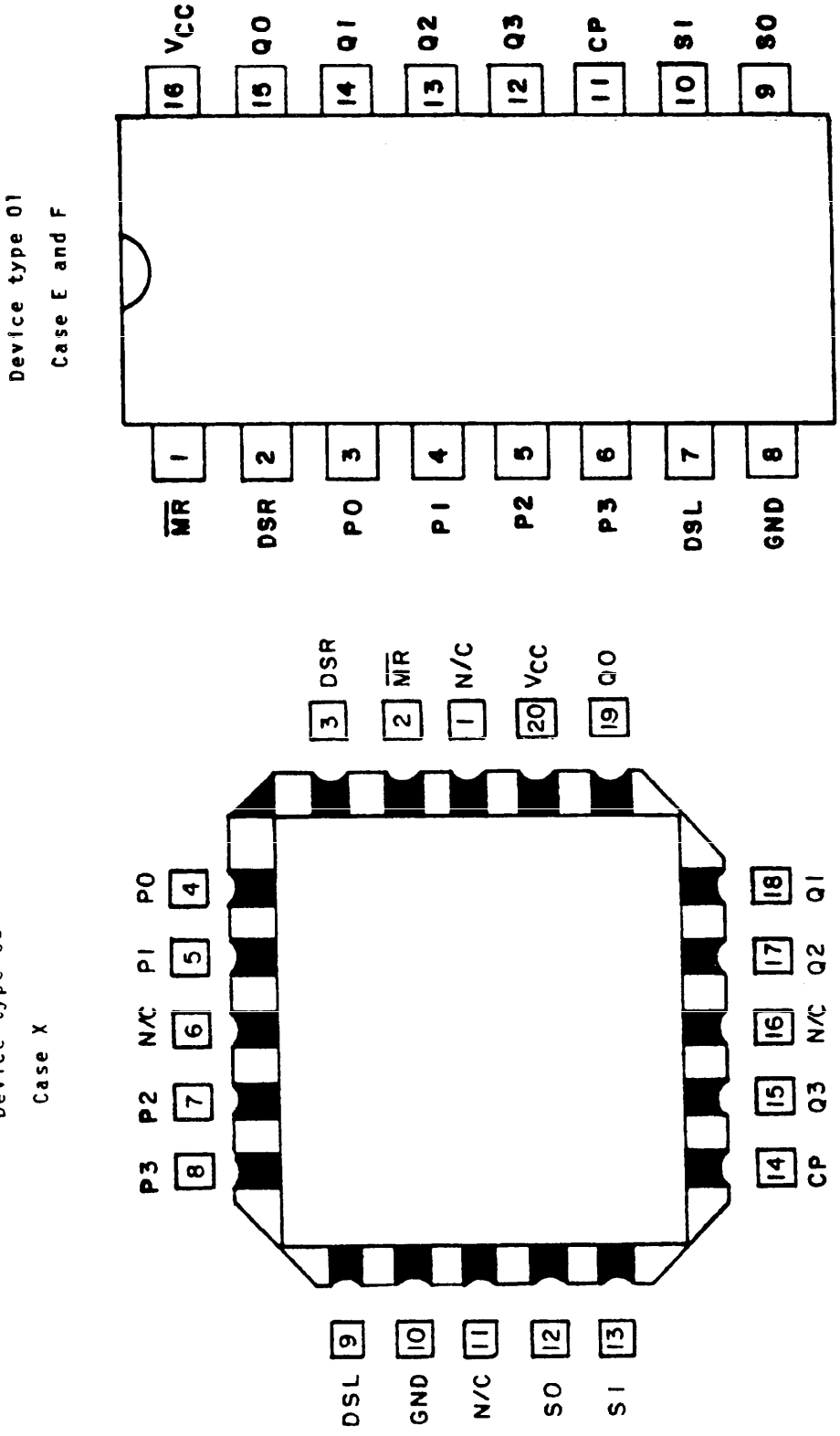



FIGURE 1. Terminal connections.



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Device type 01

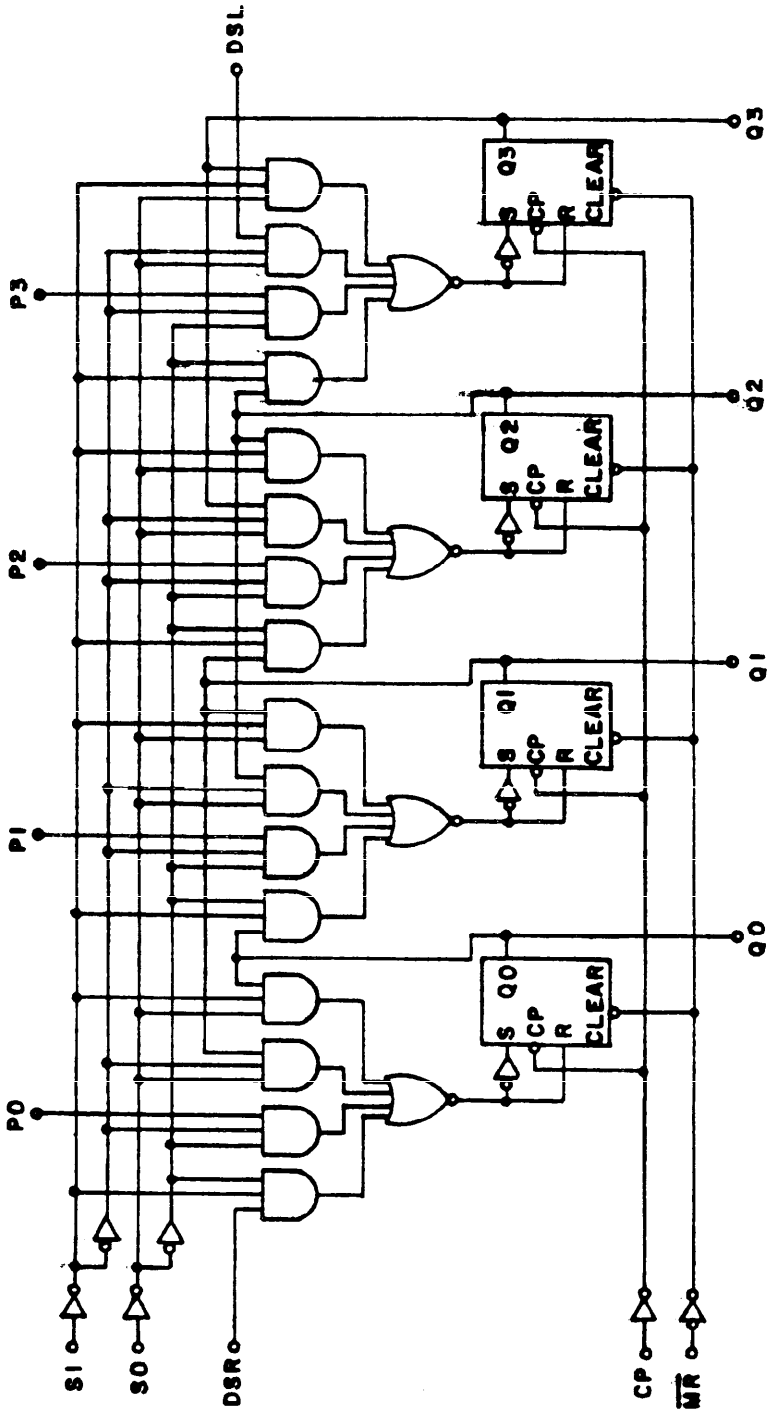



FIGURE 2. Logic diagram.



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Device type 01

| OPERATING MODE | INPUTS | | | | | | OUTPUTS | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | $\overline{MR}$ | S1 | S0 | DSR | DSL | Pn | Q0 | Q1 | Q2 | Q3 |
| Reset | L | X | X | X | X | X | L | L | L | L |
| Hold | H | I | I | X | X | X | q0 | q1 | q2 | q3 |
| Shift Left | H | h | I | X | I | X | q1 | q2 | q3 | L |
| | H | h | I | X | h | X | q1 | q2 | q3 | H |
| Shift Right | H | I | h | I | X | X | L | q0 | q1 | q2 |
| | H | I | h | h | X | X | H | q0 | q1 | q2 |
| Parallel Load | H | h | h | X | X | Pn | P0 | P1 | P2 | P3 |

I = LOW voltage level one setup time prior to the LOW-to-HIGH clock transition.

h = HIGH voltage level one setup time prior to the LOW-to-HIGH clock transition.

pn (qn) = Lower case letters indicate the state of the referenced input (or output) one setup time prior to the LOW-to-HIGH clock transition.

H = HIGH voltage level

L = LOW voltage level

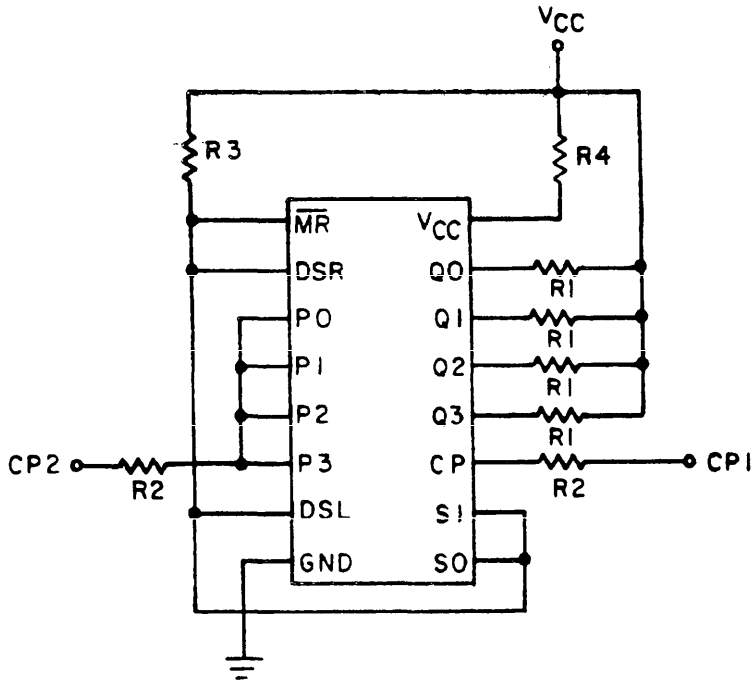
X = Immaterial

FIGURE 3. Truth table.



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DEVICE TYPE 01
CASES E, F, X
(Pins not referenced for case X are N/C)

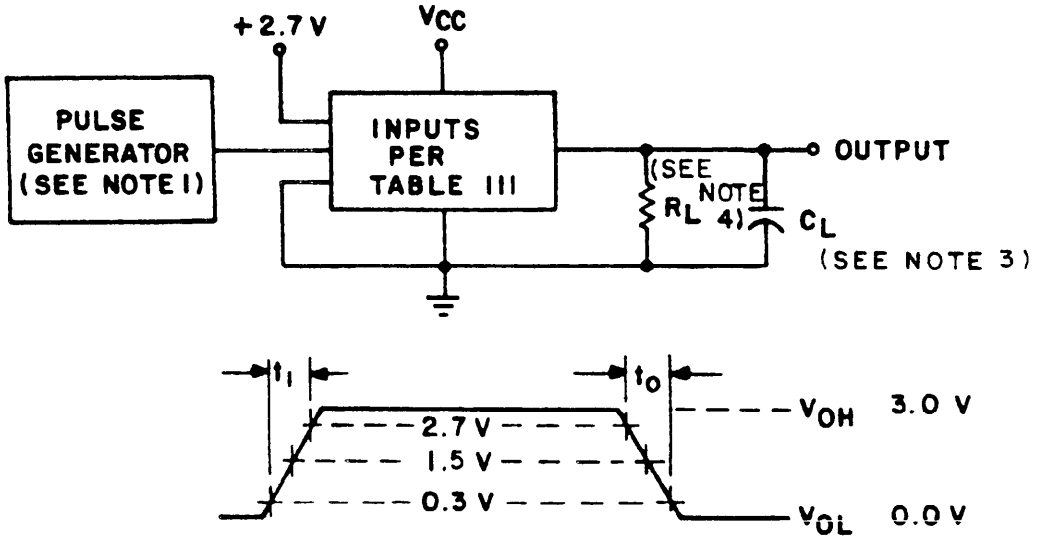



NOTES:
1. CP = 100 kHz ±50% square wave; duty cycle = 50 ±15%; CP2 = CP1 ÷ 2; $V_{IL}$ = -0.5 V minimum to 0.8 V maximum; $V_{IH}$ = 2.0 V minimum to 5.5 V maximum.
2. R1 = 240Ω ±5%; R2 = 51Ω ±5%; R3 = 1 kΩ ±5%.
3. $V_{CC}$ and R4 shall be chosen to insure 5.5 V minimum is present at device $V_{CC}$ terminal.

FIGURE 4. Burn-in and life test circuit.



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NOTES:
1. Pulse generator has the following characteristics: $t_1 = t_0 \leq 2.5$ ns, PRR $\leq$ 1 MHz, $Z_{OUT} \cong 50\Omega$.
2. Inputs not under test are at ground.
3. $C_L$ = 50 pf ±10%, including scope probe, wiring and stray capacitance without package in test fixture.
4. RL = 499Ω ±5%.
5. Voltage measurements are to be made with respect to network ground terminal.

FIGURE 5. Switching time waveform.



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Device type 01

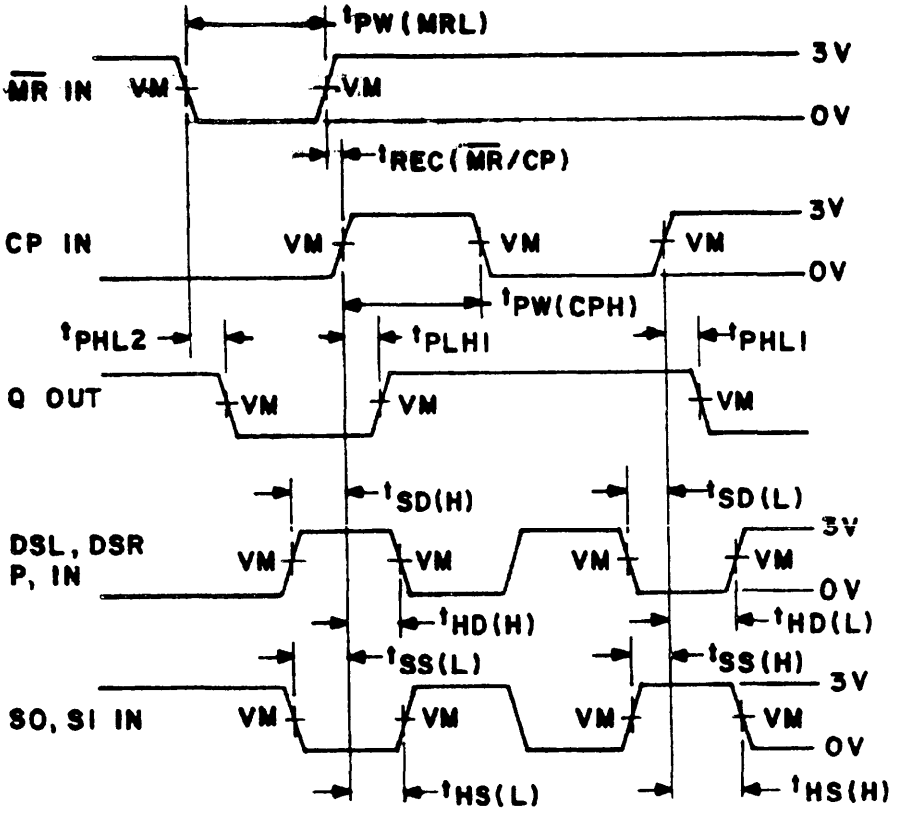



FIGURE 5. Switching time waveforms - Continued.



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Caution: The resultant power factor must be within the wattmeter operating

TABLE III. Group A inspection for device type 01.

Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open)

| Subgroup | Symbol | MIL-STD-883 method | Cases E,F / Case X 1/ / Test no. | 1 / 2 / MR | 2 / 3 / DSR | 3 / 4 / PO | 4 / 5 / P1 | 5 / 7 / P2 | 6 / 8 / P3 | 7 / 9 / DSL | 8 / 10 / GND | 9 / 12 / SO | 10 / 13 / S1 | 11 / 14 / CP | 12 / 15 / Q3 | 13 / 17 / Q2 | 14 / 18 / Q1 | 15 / 19 / Q0 | 16 / 20 / $V_{CC}$ | Measured terminal | Test limits Min | Test limits Max | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 $T_C$ = 25°C | $V_{OH}$ | 3006 | 1 | 2/ | 2.0 V | 0.8 V | 0 V | 0 V | 0 V | 0 V | GND | 2.0 V | 0.8 V | 2/ | | | | -1 mA | 4.5 V | Q0 | 2.5 | | V |
| | | " | 2 | " | 0.8 V | 2.0 V | 0 V | " | " | " | " | " | 2.0 V | " | | | | -1 mA | " | Q0 | " | | " |
| | | " | 3 | " | 0 V | " | 2.0 V | 2.0 V | 2.0 V | " | " | " | " | " | | | | | " | Q1 | " | | " |
| | | " | 4 | " | " | " | 0 V | " | 0 V | " | " | " | " | " | | | | | " | Q2 | " | | " |
| | | " | 5 | " | " | " | " | 2.0 V | " | 2.0 V | " | " | " | " | -1 mA | -1 mA | -1 mA | | " | Q3 | " | | " |
| | | " | 6 | " | " | " | " | 0 V | " | " | " | 0.8 V | " | " | -1 mA | | | | " | Q3 | " | | " |
| | $V_{OL}$ | 3007 | 7 | 5.5 V | 0.8 V | 2.0 V | " | 0.8 V | " | 0 V | " | 2.0 V | 0.8 V | 3/ | | | | 20 mA | " | Q0 | | 0.5 | " |
| | | " | 8 | " | 2.0 V | 0.8 V | 0.8 V | 0 V | " | " | " | " | " | " | | | | 20 mA | " | Q0 | | " | " |
| | | " | 9 | " | 0 V | " | 0 V | 0 V | 0.8 V | " | " | " | " | " | | | | | " | Q1 | | " | " |
| | | " | 10 | " | " | " | " | 0.8 V | 2.0 V | 2.0 V | " | " | " | " | | | | | " | Q2 | | " | " |
| | | " | 11 | " | " | " | " | 0 V | " | 0.8 V | " | " | " | " | 20 mA | 20 mA | 20 mA | | " | Q3 | | " | " |
| | | " | 12 | " | " | " | " | " | " | " | " | 0.8 V | " | " | 20 mA | | | | " | Q3 | | " | " |
| | $V_{IC}$ | | 13 | -18 mA | -18 mA | | | | | | | 0 V | 5.5 V | | | | | | | MR | | -1.2 | |
| | | | 14 | | | -18 mA | | | | | | " | 0 V | | | | | | | DSR | | " | |
| | | | 15 | | | | -18 mA | | | | | " | " | | | | | | | PO | | " | |
| | | | 16 | | | | | -18 mA | | | | " | " | | | | | | | P1 | | " | |
| | | | 17 | | | | | | -18 mA | | | " | " | | | | | | | P2 | | " | |
| | | | 18 | | | | | | | -18 mA | | " | " | | | | | | | P3 | | " | |
| | | | 19 | | | | | | | | | 5.5 V | -18 mA | | | | | | | DSL | | " | |
| | | | 20 | | | | | | | | | -18 mA | | -18 mA | | | | | | SO | | " | |
| | | | 21 | | | | | | | | | | | | | | | | | S1 | | " | |
| | | | 22 | | | | | | | | | | | | | | | | | CP | | " | |
| | $I_{IH1}$ | 3010 | 23 | 2.7 V | | | | | | | | 0 V | 5.5 V | | | | | | 5.5 V | MR | | 20 | μA |
| | | " | 24 | | 2.7 V | | | | | | | " | 0 V | | | | | | " | DSR | | " | " |
| | | " | 25 | | | 2.7 V | | | | | | " | " | | | | | | " | PO | | " | " |
| | | " | 26 | | | | 2.7 V | | | | | " | " | | | | | | " | P1 | | " | " |
| | | " | 27 | | | | | 2.7 V | | | | " | " | | | | | | " | P2 | | " | " |
| | | " | 28 | | | | | | 2.7 V | | | " | " | | | | | | " | P3 | | " | " |
| | | " | 29 | | | | | | | 2.7 V | | " | " | | | | | | " | DSL | | " | " |
| | | " | 30 | | | | | | | | | 5.5 V | 2.7 V | | | | | | " | SO | | " | " |
| | | " | 31 | | | | | | | | | 2.7 V | | 2.7 V | | | | | " | S1 | | " | " |
| | | " | 32 | | | | | | | | | | | | | | | | " | CP | | " | " |
| | $I_{IH2}$ | " | 33 | 7.0 V | | | | | | | | 0 V | 5.5 V | | | | | | | MR | | 100 | |
| | | " | 34 | | 7.0 V | | | | | | | " | 0 V | | | | | | | DSR | | " | |
| | | " | 35 | | | 7.0 V | | | | | | " | " | | | | | | | PO | | " | |
| | | " | 36 | | | | 7.0 V | | | | | " | " | | | | | | | P1 | | " | |
| | | " | 37 | | | | | 7.0 V | | | | " | " | | | | | | | P2 | | " | |
| | | " | 38 | | | | | | 7.0 V | | | " | " | | | | | | | P3 | | " | |
| | | " | 39 | | | | | | | 7.0 V | | " | " | | | | | | | DSL | | " | |
| | | " | 40 | | | | | | | | | 5.5 V | 7.0 V | | | | | | | SO | | " | |
| | | " | 41 | | | | | | | | | 7.0 V | | 7.0 V | | | | | | S1 | | " | |
| | | " | 42 | | | | | | | | | | | | | | | | | CP | | " | |
| | $I_{IL1}$ | 3009 | 43 | 0.5 V | | | | | | | | 5.5 V | 0 V | | | | | | | MR | 4/ | 4/ | mA |
| | | " | 44 | | 0.5 V | | | | | | | " | 5.5 V | | | | | | | DSR | " | " | " |
| | | " | 45 | | | 0.5 V | | | | | | " | " | | | | | | | PO | " | " | " |
| | | " | 46 | | | | 0.5 V | | | | | " | " | | | | | | | P1 | " | " | " |
| | | " | 47 | | | | | 0.5 V | | | | " | " | | | | | | | P2 | " | " | " |
| | | " | 48 | | | | | | 0.5 V | | | " | " | | | | | | | P3 | " | " | " |
| | | " | 49 | | | | | | | 0.5 V | | " | " | | | | | | | DSL | " | " | " |
| | | " | 50 | | | | | | | | | 0 V | 0.5 V | | | | | | | SO | " | " | " |
| | | " | 51 | | | | | | | | | 0.5 V | | 0.5 V | | | | | | S1 | " | " | " |
| | | " | 52 | | | | | | | | | | | | | | | | | CP | " | " | " |

See footnotes at end of table.





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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open)

| Subgroup | Symbol | MIL-STD-883 method | Cases E,F Case X 1/ Test no. | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | Measured terminal | Test limits Min | Test limits Max | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | 2 | 3 | 4 | 5 | 7 | 8 | 9 | 10 | 12 | 13 | 14 | 15 | 17 | 18 | 19 | 20 | | | | |
| | | | | WR | OSR | PO | P1 | P2 | P3 | DSL | GND | SO | S1 | CP | Q3 | Q2 | Q1 | Q0 | $V_{CC}$ | | | | |
| 1<br>$T_C$ = 25°C | $I_{OS}$ | 3011 | 53 | 5.5 V | | 5.5 V | 0 V | 0 V | 0 V | | GND | 5.5 V | 5.5 V | 3/ | | | | 0 V | 5.5 V | Q0 | -60 | -150 | mA |
| | | " | 54 | " | | 0 V | 5.5 V | 0 V | " | | " | " | " | " | | | | | " | Q1 | " | " | " |
| | | " | 55 | " | | " | 0 V | 5.5 V | 5.5 V | | " | " | " | " | | 0 V | | | " | Q2 | " | " | " |
| | | " | 56 | " | | " | 0 V | 0 V | " | | " | " | " | " | 0 V | | 0 V | | " | Q3 | " | " | " |
| | $I_{OD}$ | | 57 | 0 V | | | | | | | " | | | | 2.5 V | | | | 4.5 V | Q3 | 60 | | " |
| | | | 58 | " | | | | | | | " | | | | | 2.5 V | | | " | Q2 | " | | " |
| | | | 59 | " | | | | | | | " | | | | | | 2.5 V | | " | Q1 | " | | " |
| | | | 60 | " | | | | | | | " | | | | | | | 2.5 V | " | Q0 | " | | " |
| | $I_{CC}$ | 3005 | 61 | 5.5 V | 5.5 V | 0 V | 0 V | 0 V | 0 V | 5.5 V | " | 5.5 V | 5.5 V | 3/ | | | | | 5.5 V | $V_{CC}$ | | 46 | " |
| 2 | Same tests, terminal conditions, and limits as for subgroup 1, except $T_C$ = +125°C and $V_{IC}$ tests are omitted. | | | | | | | | | | | | | | | | | | | | | | |
| 3 | Same tests, terminal conditions, and limits as for subgroup 1, except $T_C$ = -55°C and $V_{IC}$ tests are omitted. | | | | | | | | | | | | | | | | | | | | | | |
| 7<br>$T_C$ = 25°C | Functional tests 5/ | 3014 | 62 | B | A | A | A | A | A | A | GND | A | A | A | L | L | L | L | 6/ | All outputs | | | |
| | | " | 63 | A | A | " | " | " | " | " | " | " | " | A | H | H | H | H | " | | | | |
| | | " | 64 | " | B | " | " | " | " | " | " | " | " | B | L | L | H | H | " | | | | |
| | | " | 65 | " | B | " | " | " | " | " | " | " | " | A | H | H | L | L | " | | | | |
| | | " | 66 | " | A | B | B | B | B | B | " | " | " | B | L | H | H | L | " | | | | |
| | | " | 67 | " | A | B | B | B | B | A | " | " | " | A | H | L | H | L | " | | | | |
| | | " | 68 | " | B | A | B | A | A | B | " | " | " | B | L | H | L | H | " | | | | |
| | | " | 69 | " | B | A | A | A | A | A | " | " | " | A | H | L | L | H | " | | | | |
| | | " | 70 | " | A | B | A | B | B | A | " | " | " | B | L | H | H | L | " | | | | |
| | | " | 71 | " | " | " | " | " | " | " | " | " | " | A | H | L | H | L | " | | | | |
| | | " | 72 | " | " | " | " | " | " | " | " | " | " | B | L | H | L | H | " | | | | |
| | | " | 73 | " | " | " | " | " | " | " | " | " | " | A | H | L | L | H | " | | | | |
| | | " | 74 | " | " | " | " | " | " | " | " | " | " | B | L | H | H | L | " | | | | |
| | | " | 75 | " | B | " | " | " | " | " | " | " | " | A | H | L | H | L | " | | | | |
| | | " | 76 | " | " | " | " | " | " | " | " | " | " | B | L | H | L | H | " | | | | |
| | | " | 77 | " | " | " | " | " | " | " | " | " | " | A | H | L | L | H | " | | | | |
| | | " | 78 | " | " | " | " | " | " | " | " | " | " | B | L | H | H | L | " | | | | |
| | | " | 79 | " | " | " | " | " | " | " | " | " | " | A | H | L | H | L | " | | | | |
| | | " | 80 | " | " | " | " | " | " | " | " | " | " | B | L | H | L | H | " | | | | |
| | | " | 81 | " | " | " | " | " | " | " | " | " | " | A | H | L | L | H | " | | | | |
| | | " | 82 | " | " | " | " | " | " | " | " | " | " | B | L | H | H | L | " | | | | |
| | | " | 83 | " | " | " | " | " | " | " | " | " | " | A | H | L | H | L | " | | | | |
| | | " | 84 | " | " | " | " | " | " | " | " | " | " | B | L | H | L | H | " | | | | |
| | | " | 85 | " | " | A | " | " | " | " | " | " | " | A | H | L | L | H | " | | | | |
| | | " | 86 | " | " | " | A | A | A | B | " | " | " | B | L | H | H | L | " | | | | |
| | | " | 87 | " | " | " | " | " | " | " | " | " | " | A | H | L | H | L | " | | | | |
| | | " | 88 | " | " | " | " | " | " | " | " | " | " | B | L | H | L | H | " | | | | |
| | | " | 89 | " | " | " | " | " | " | " | " | " | " | A | H | L | L | H | " | | | | |
| | | " | 90 | " | " | " | " | " | " | " | " | " | " | B | L | H | H | L | " | | | | |
| | | " | 91 | " | " | " | " | " | " | " | " | " | " | A | H | L | H | L | " | | | | |
| | | " | 92 | " | " | " | " | " | " | " | " | " | " | B | L | H | L | H | " | | | | |
| | | " | 93 | " | " | " | " | " | " | " | " | " | " | A | H | L | L | H | " | | | | |
| | | " | 94 | " | " | " | " | " | " | " | " | " | " | B | L | H | H | L | " | | | | |
| | | " | 95 | " | " | " | " | " | " | " | " | " | " | A | H | L | H | L | " | | | | |
| | | " | 96 | " | " | " | " | " | " | " | " | " | " | B | L | H | L | H | " | | | | |
| | | " | 97 | " | " | " | " | " | " | " | " | " | " | A | H | L | L | H | " | | | | |
| | | " | 98 | B | " | " | " | " | " | " | " | " | " | B | L | H | H | L | " | | | | |
| | | " | 99 | A | " | " | " | " | " | " | " | " | " | A | H | L | H | L | " | | | | |
| | | " | 100 | " | " | " | " | " | " | " | " | " | " | B | L | H | L | H | " | | | | |
| | | " | 101 | " | " | B | " | B | B | " | " | " | " | A | H | L | L | H | " | | | | |
| | | " | 102 | " | " | " | B | " | " | " | " | " | " | B | L | H | H | L | " | | | | |
| | | " | 103 | " | " | " | " | " | " | " | " | " | " | A | H | L | H | L | " | | | | |
| | | " | 104 | " | " | " | " | " | " | " | " | " | " | B | L | H | L | H | " | | | | |
| | | " | 105 | " | A | " | " | " | " | A | " | " | " | A | H | L | L | H | " | | | | |
| | | " | 106 | " | " | " | " | " | " | " | " | " | " | B | L | H | H | L | " | | | | |
| | | " | 107 | " | " | " | " | " | " | " | " | " | " | A | H | L | H | L | " | | | | |
| | | " | 108 | " | " | A | A | A | A | " | " | " | " | B | L | H | L | H | " | | | | |
| | | " | 109 | " | " | " | " | " | " | " | " | " | " | A | H | L | L | H | " | | | | |
| | | " | 110 | " | " | " | " | " | " | " | " | " | " | A | H | L | H | L | " | | | | |

See footnotes at end of table.





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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open)

| Subgroup | Symbol | MIL-STD-883 method | Cases E,F Case X 1/ Test no. | 1 / 2 / MR | 2 / 3 / DSR | 3 / 4 / P0 | 4 / 5 / P1 | 5 / 7 / P2 | 6 / 8 / P3 | 7 / 9 / DSL | 8 / 10 / GND | 9 / 12 / S0 | 10 / 13 / S1 | 11 / 14 / CP | 12 / 15 / Q3 | 13 / 17 / Q2 | 14 / 18 / Q1 | 15 / 19 / Q0 | 16 / 20 / $V_{CC}$ | Measured terminal | Test limits Min | Test limits Max | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 8 | Repeat subgroup 7 tests at $T_C$ = +125°C and $T_C$ = -55°C. | | | | | | | | | | | | | | | | | | | | | |
| 9 $T_C$ = 25°C | $t_{PLH1}$ | 3003 | 111 | 2.7 V | | 2.7 V | 2.7 V | 2.7 V | 2.7 V | | GND | 2.7 V | 2.7 V | IN | OUT | OUT | OUT | OUT | 5.0 V | CP to Q0 | 3.5 | 7.0 | ns |
| | | " | 112 | " | | | | | | | " | " | " | " | | | | | " | CP to Q1 | " | " | " |
| | | " | 113 | " | | | | | | | " | " | " | " | | | | | " | CP to Q2 | " | " | " |
| | | " | 114 | " | | | | | | | " | " | " | " | | | | | " | CP to Q3 | " | " | " |
| | $t_{PHL1}$ | " | 115 | " | | 0 V | 0 V | 0 V | 0 V | | " | " | " | " | OUT | OUT | OUT | OUT | " | CP to Q0 | " | " | " |
| | | " | 116 | " | | | | | | | " | " | " | " | | | | | " | CP to Q1 | " | " | " |
| | | " | 117 | " | | | | | | | " | " | " | " | | | | | " | CP to Q2 | " | " | " |
| | | " | 118 | " | | | | | | | " | " | " | " | | | | | " | CP to Q3 | " | " | " |
| | $t_{PHL2}$ | " | 119 | IN | | 2.7 V | 2.7 V | 2.7 V | 2.7 V | | " | " | " | | OUT | OUT | OUT | OUT | " | $\overline{MR}$ to Q0 | 4.5 | 12.0 | " |
| | | " | 120 | " | | | | | | | " | " | " | | | | | | " | $\overline{MR}$ to Q1 | " | " | " |
| | | " | 121 | " | | | | | | | " | " | " | | | | | | " | $\overline{MR}$ to Q2 | " | " | " |
| | | " | 122 | " | | | | | | | " | " | " | | | | | | " | $\overline{MR}$ to Q3 | " | " | " |
| 10 | $t_{PLH1}$ | Same tests and terminal conditions as for subgroup 9, except $T_C$ = +125°C and limits as shown. | | | | | | | | | | | | | | | | | | | 3.0 | 8.5 | " |
| | $t_{PHL1}$ | | | | | | | | | | | | | | | | | | | | 3.0 | 8.5 | " |
| | $t_{PHL2}$ | | | | | | | | | | | | | | | | | | | | 4.5 | 14.5 | " |
| 11 | Same tests, terminal conditions, and limits as for subgroup 10, except $T_C$ = -55°C. | | | | | | | | | | | | | | | | | | | | | | |

1/ For case X pins not referenced are N/C.

2/ Apply 3 V, 0 V, 3 V to $\overline{MR}$ then 3 V, 0 V, 3 V to CP prior to making measurement.

3/ Apply 3 V, 0 V, 3 V to CP prior to making measurement.

4/ $I_{IL}$ limits shall be as follows:

| Parameter | Test no. | Limits in mA for circuits A | B | C |
|---|---|---|---|---|
| $I_{IL1}$ | 43 through 52 | -.25/-.60 | -.03/-.60 | -.03/-.60 |

5/ A = 4.5 V, B = 0.0 V, X = don't care
H ≥2.5 V, L ≤0.5 V with high speed checker double comparator, or
H ≥1.5 V, L ≤1.5 V with high speed checker single comparator.

6/ Perform function sequence at $V_{CC}$ = 4.5 V and repeat at $V_{CC}$ = 5.5 V.





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c. Steady state life test (method 1005 of MIL-STD-883) conditions:

(1) Test condition D or E, using the circuit shown on figure 4 or equivalent.

(2) $T_A$ = +125°C minimum.

(3) Test duration: 1,000 hours, except as permitted by appendix B of MIL-M-38510.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883. End-point electrical parameters shall be as specified in table II herein.

4.5 Methods of inspection. Methods of inspection shall be specified as follows:

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit ground terminal. Currents given are conventional and positive when flowing into the referenced terminal.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be as specified in MIL-M-38510.

6. NOTES

6.1 Notes. The notes specified in MIL-M-38510 are applicable to this specification.

6.2 Intended use. Microcircuits conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.3 Ordering data. The acquisition document should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirements for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to the contracting activity in addition to notification to the qualifying activity, if applicable.

e. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action, and reporting of results, if applicable.

f. Requirements for product assurance options.

g. Requirements for special carriers, lead lengths, or lead forming, if applicable. These requirements shall not affect the part number. Unless otherwise specified, these requirements shall not apply to direct purchase by or direct shipment to the Government.

h. Requirements for "JAN" marking.

6.4 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

| | |
|---|---|
| GND | Ground zero voltage potential |
| $V_{IN}$ | Voltage level at an input terminal |
| $I_{IN}$ | Current flowing into an input terminal |



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6.5 Logistic support. Lead materials and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2), lead material and finish C (see 3.3). Longer lead lengths and lead forming shall not affect the part number.

6.6 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

| Military device type | Generic-industry type |
|---|---|
| 01 | 54F194 |

6.7 Manufacturers' designations. Manufacturers' circuits which form a part of this specification are designated with an "X" as shown in table IV herein.

TABLE IV. Manufacturers' designation.

| | Circuits | | |
|---|---|---|---|
| Device type | A | B | C |
| | Fairchild | Motorola | Signetics |
| 01 | X | | |

6.8 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Navy - OS, SH
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

Agent:
DLA - ES

Preparing activity:
Air Force - 17

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*(See Instructions – Reverse Side)*

| 1. DOCUMENT NUMBER | 2. DOCUMENT TITLE |
|---|---|
| MIL-M-38510/336A | MICROCIRCUITS, DIGITAL, ADVANCED SCHOTTKY TTL, SHIFT REGISTERS, MONOLITHIC SILICON |

| 3a. NAME OF SUBMITTING ORGANIZATION | 4. TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| | ☐ VENDOR |
| | ☐ USER |
| b. ADDRESS *(Street, City, State, ZIP Code)* | ☐ MANUFACTURER |
| | ☐ OTHER *(Specify)*: _____ |

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

| 7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional | b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional |
|---|---|
| c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional | 8. DATE OF SUBMISSION *(YYMMDD)* |

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